

In search of explosive primes: the inverse volcano problem

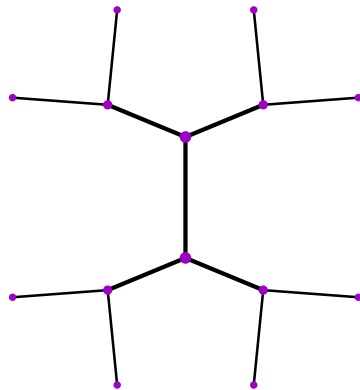
Max Kortge @ University of Melbourne
(joint with Alex Ghitza + Dhruv Gupta)

1 September 2026, NTDU14

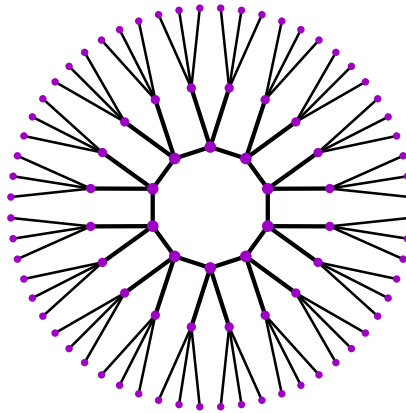
Volcanoes!

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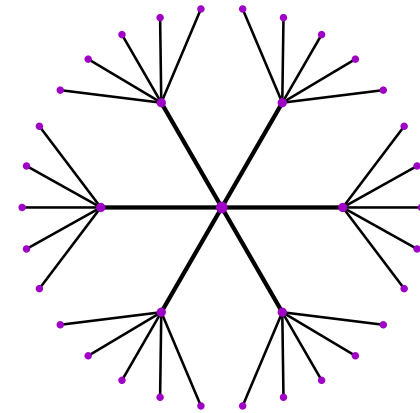
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$(R_2, 2, 2)$



$(S_{10}, 3, 2)$

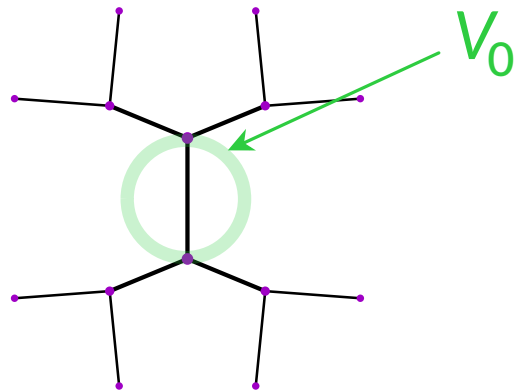


$(I_1, 5, 2)$

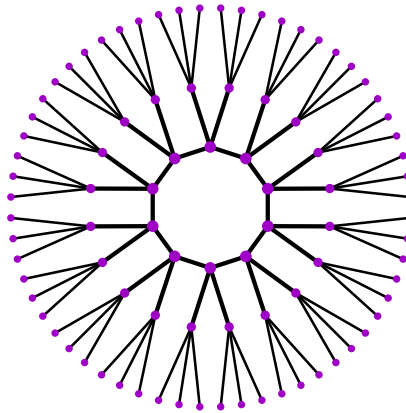
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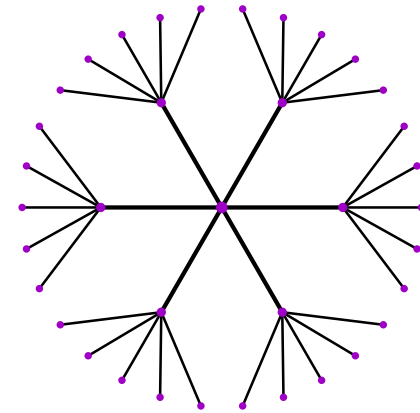
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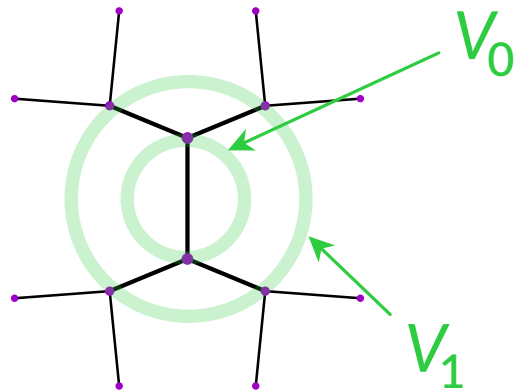


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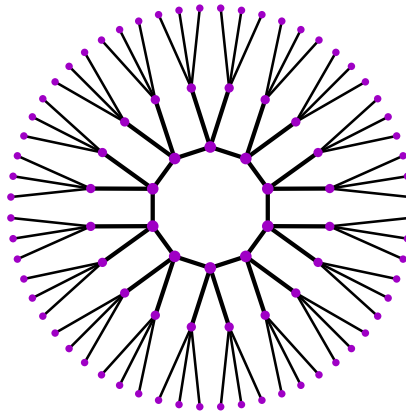
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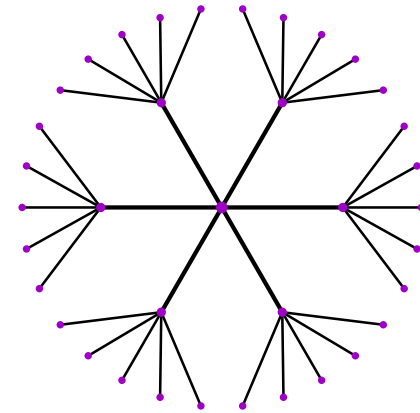
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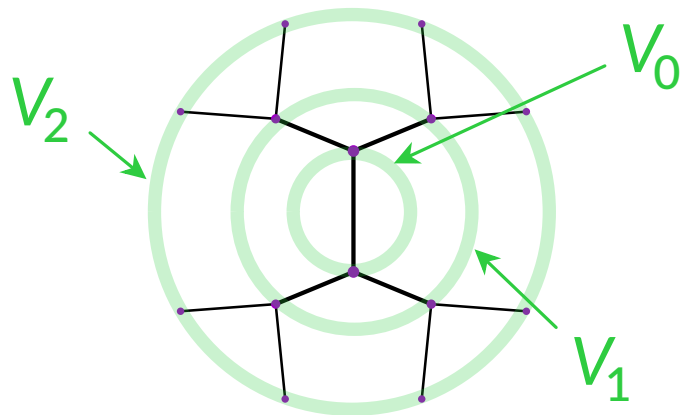


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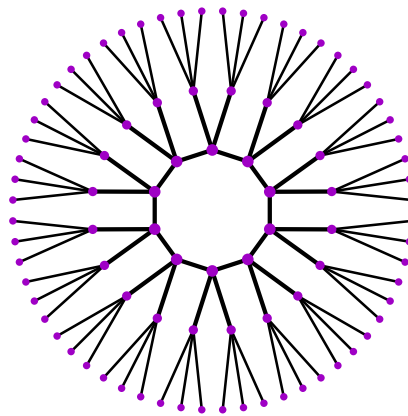
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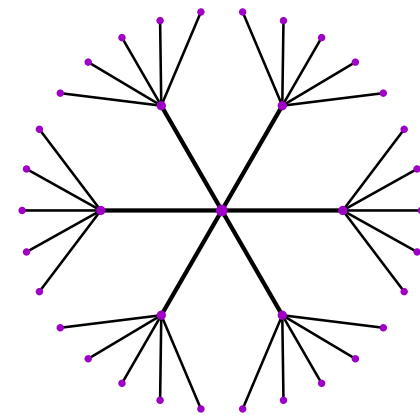
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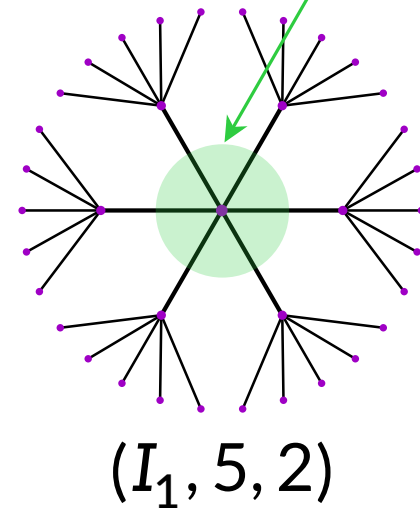
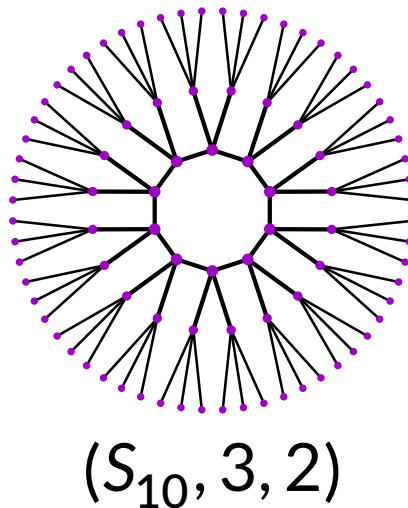
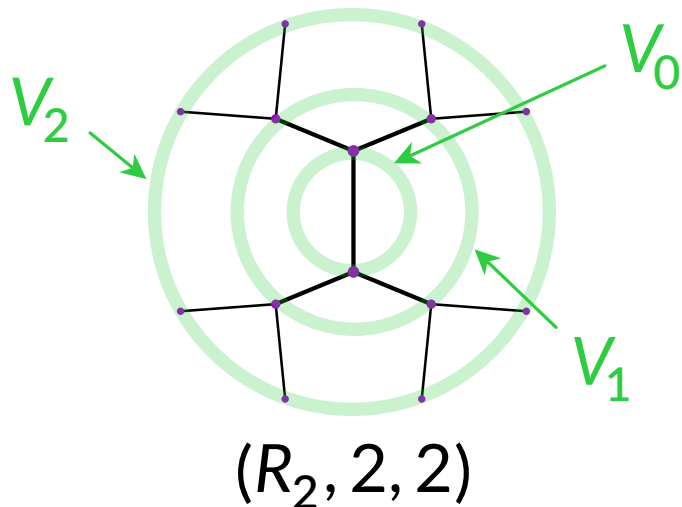


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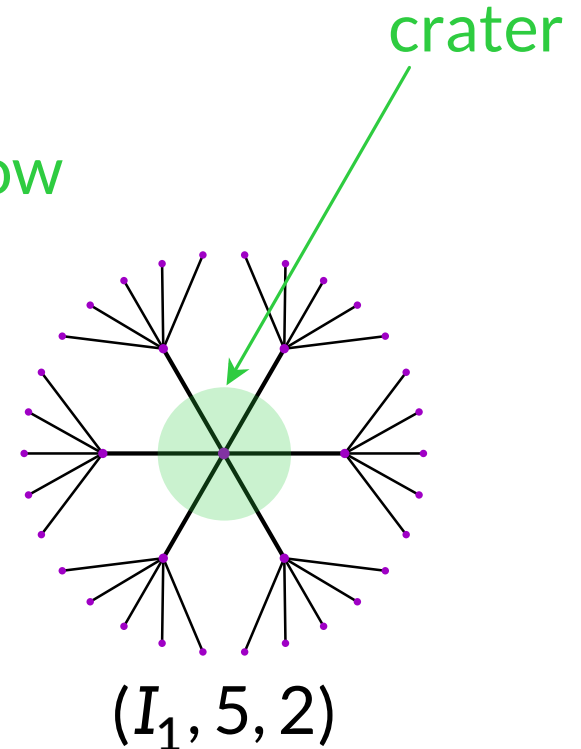
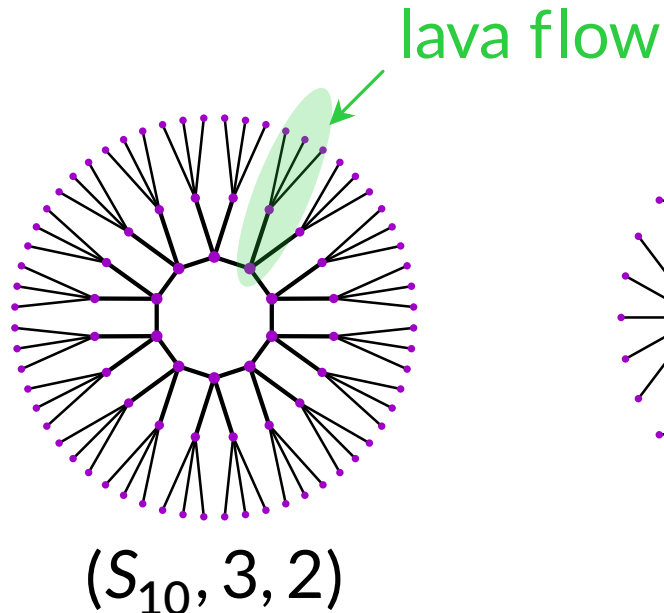
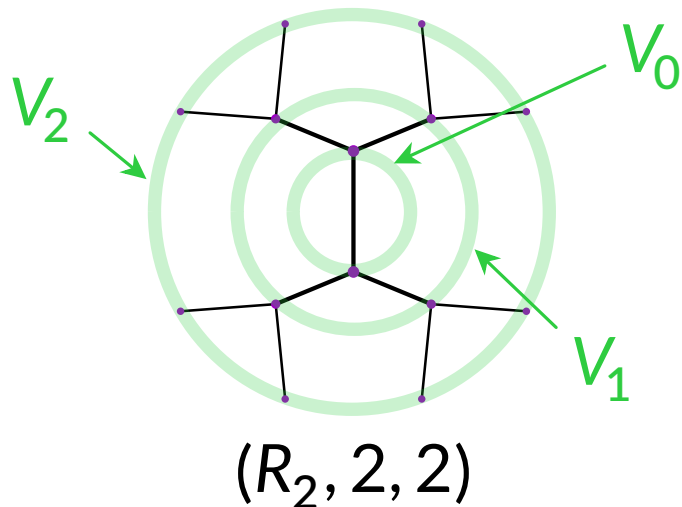


crater

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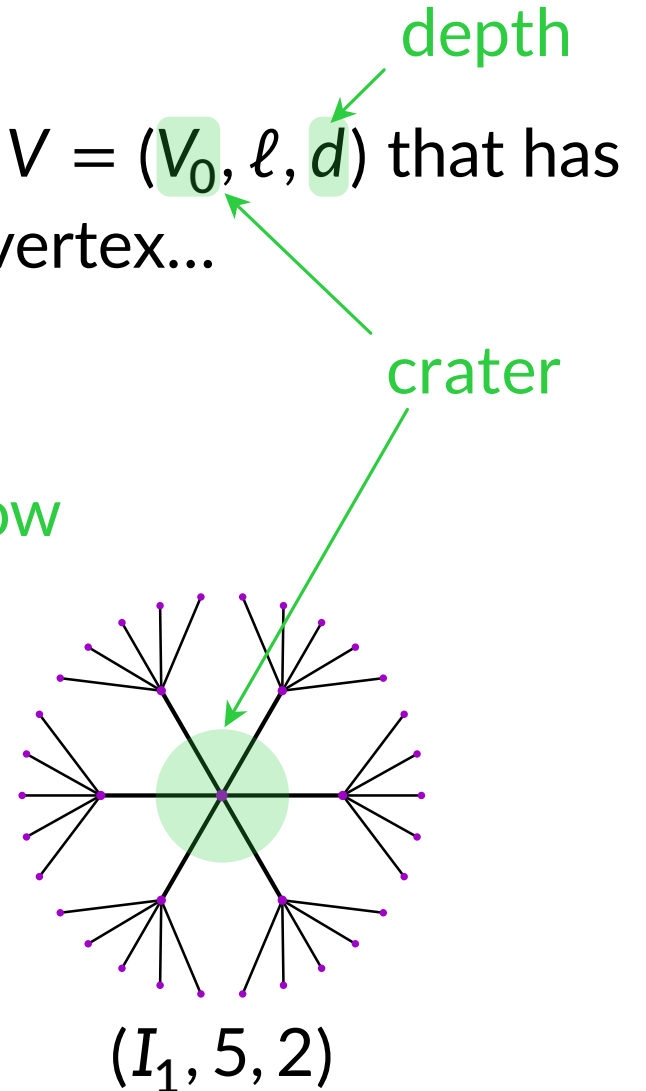
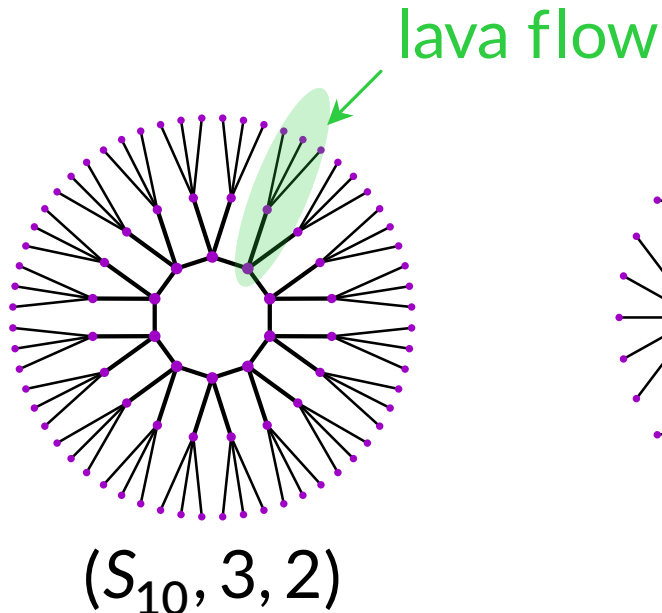
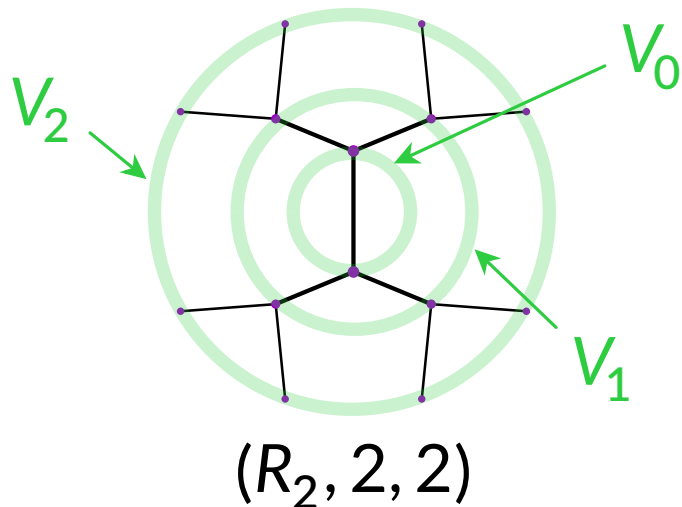
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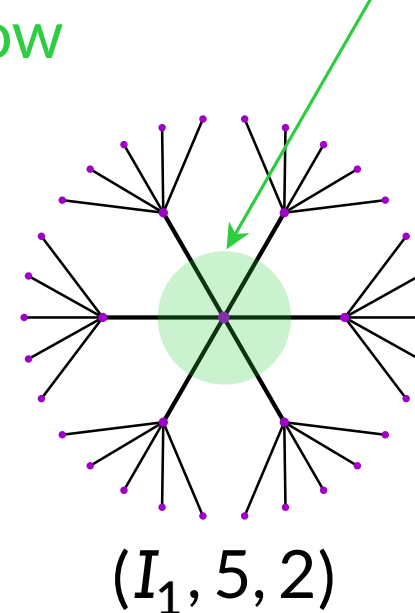
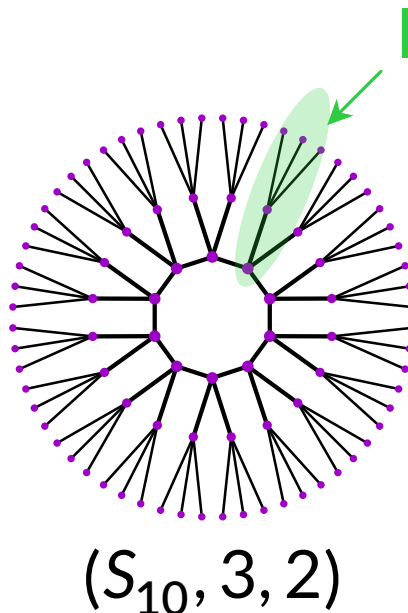
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<https://spotlight.unavco.org/station-pages/puoc/puoc.html>



depth

crater

lava flow

Why?

Consider $E/\mathbb{F}_{p^k}$ an elliptic curve. E is **ordinary** if $\text{End}(E)$ is an order $\mathcal{O}$ in an imaginary quadratic field K . (Other possibility is **supersingular**.)

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For a prime $\ell \neq p$, the **ℓ -isogeny graph** $\mathcal{G}_\ell(\mathbb{F}_{p^k})$ has:

- ▶ vertices $\{j \in \mathbb{F}_{p^k} : j \text{ ordinary, } j \neq 0, 1728\},$
- ▶ an edge from $j(E_1)$ to $j(E_2)$ if there is an ℓ -isogeny $\varphi : E_1 \rightarrow E_2.$

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Existence of dual isogeny $\Rightarrow$ undirected.

Story thus far

Theorem (Kohel). Every connected component of $\mathcal{G}_\ell(\mathbb{F}_{p^k})$ is an ℓ -volcano. (Awesome name is thanks to Fouquet–Morain.)

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We sought to understand and extend this.

Our results: settled cases

V_0	ℓ	d	n	Condition	k -  primes
S_n, I_1	≥ 3	≥ 1	≥ 1	$r < d$	Infinitely many
	2	≥ 2	≥ 1	$r < d - 1$	
	2	1	≥ 1	$r = 0$	
R_2	≥ 2	≥ 1	—	$r < d + 1$	
R_1	$\neq 3$	≥ 1	—	$r < d + 1$	
	3	≥ 1	—	$r < d$	
S_n, I_1	2	1	≥ 1	$r \geq 1$	None
	2	2, 3	≥ 1	$r \geq d - 1$	
S_n	2	≥ 4	$\in \{1, 2, 3, 5, 6, 7\}$	$r \geq d - 1$	
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$(S_2, 2, 1)$

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Will do this case

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$(S_2, 2, 1)$

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V_0	ℓ	d	n	Condition	on $r = v_\ell(k)$ k -  primes
I_1	≥ 3	≥ 1	—	$r \geq d$	Infinitely many*
R_2	≥ 5	≥ 1	—	$r \geq d + 1$	
S_n	≥ 3	≥ 1	≥ 1	$r \geq d$	Don't know
S_n	2	≥ 4	$\notin \{1, 2, 3, 5, 6, 7\}$	$r \geq d - 1$	
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*Conditional on modified Cohen–Lenstra heuristics.

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Missing from the table is $d = 0$: we proved there are infinitely many k -explosive primes in all cases (ℓ is not fixed).

Imaginary quadratic orders

Let K be an imaginary quadratic field. An **order** is a subring $\mathcal{O}$ of finite index in the ring of integers $\mathcal{O}_K$.

- ▶ Has form $\mathcal{O} = \mathbb{Z} + c\mathcal{O}_K$ where $c = [\mathcal{O}_K : \mathcal{O}] \in \mathbb{Z}^+$ is called the **conductor** of $\mathcal{O}$.

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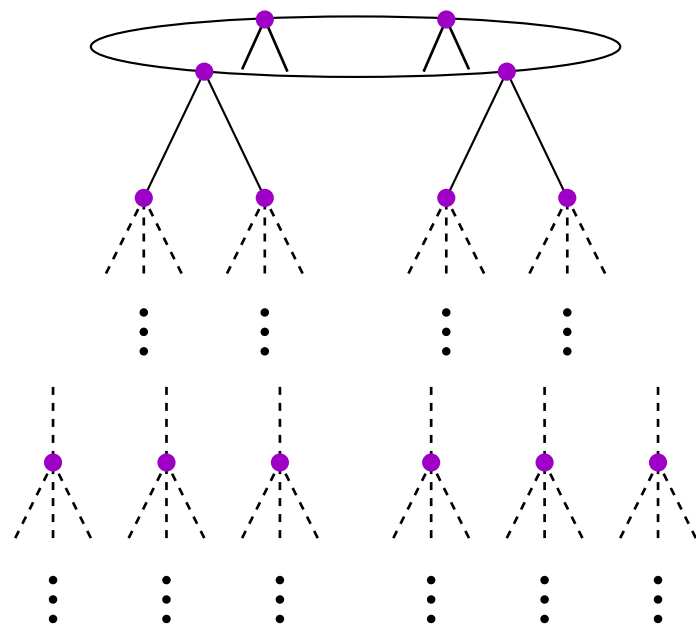
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Example. $K = \mathbb{Q}(\sqrt{-47})$.

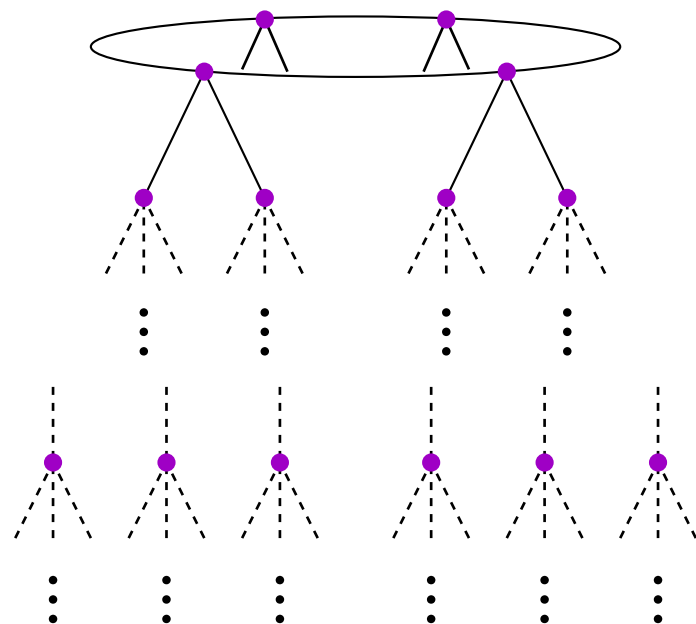
- ▶ $\mathcal{O}_K = \mathbb{Z}[(1 + \sqrt{-47})/2]$.
- ▶ The **ideal class group** is $\text{Cl}(\mathcal{O}_K) \cong \mathbb{Z}/5\mathbb{Z}$ and the **class number** is $h(\mathcal{O}_K) = 5$.
- ▶ $\mathcal{O} = \mathbb{Z}[\sqrt{-47}]$ is the order of conductor 2 and $\text{Cl}(\mathcal{O}) \cong \mathbb{Z}/5\mathbb{Z}$.

Lava flows



Level	Order		Class group
V_0	$\mathcal{O}_0$	c	$Cl(\mathcal{O}_0)$
	$\cup$		$\uparrow \pi_0$
V_1	$\mathcal{O}_1$	$c\ell$	$Cl(\mathcal{O}_1)$
	$\cup$		$\uparrow \pi_1$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
	$\cup$		$\uparrow \pi_{i-1}$
V_i	$\mathcal{O}_i$	$c\ell^i$	$Cl(\mathcal{O}_i)$
	$\cup$		$\uparrow \pi_i$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Lava flows



Level

Order

conductor coprime to ℓ

Class group

V_0

$\mathcal{O}_0$

c

$\text{Cl}(\mathcal{O}_0)$

V_1

$\mathcal{O}_1$

$c\ell$

$\text{Cl}(\mathcal{O}_1)$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

V_i

$\mathcal{O}_i$

$c\ell^i$

$\text{Cl}(\mathcal{O}_i)$

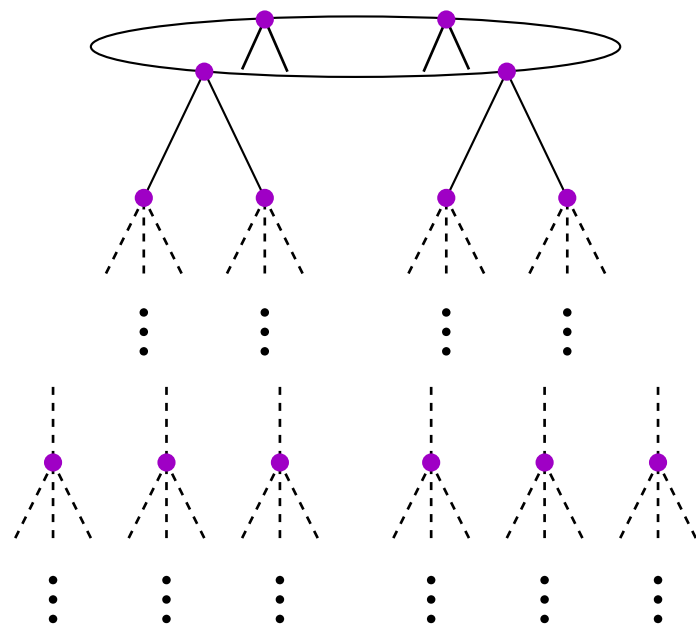
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$\vdots$

$\vdots$

$\vdots$

Lava flows



Level

Order

Class group

V_0

$\mathcal{O}_0$

c

$\text{Cl}(\mathcal{O}_0)$

V_1

$\mathcal{O}_1$

$c\ell$

$\text{Cl}(\mathcal{O}_1)$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

V_i

$\mathcal{O}_i$

$c\ell^i$

$\text{Cl}(\mathcal{O}_i)$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

conductor coprime to ℓ

$$\forall p \subseteq \mathcal{O}_K, [p \cap \mathcal{O}_{i+1}] \mapsto [p \cap \mathcal{O}_i]$$

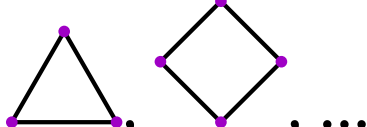
Crater

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Let $\mathcal{L} \subseteq \mathcal{O}_0$ be a prime ideal above ℓ .

Type of V_0	ℓ in $\mathcal{O}_0$	$\text{ord}([\mathcal{L}])$ in $\text{Cl}(\mathcal{O}_0)$	Picture
I_1	inert	1	
R_1	ramified	1	
S_1	split	1	
R_2	ramified	2	
S_2	split	2	
$S_n, n \geq 3$	split	n	

Crater cont.

Example. $K = \mathbb{Q}(\sqrt{-47})$, $\mathcal{O}_K = \mathbb{Z}[(1 + \sqrt{-47})/2]$, $\text{Cl}(\mathcal{O}_K) \cong \mathbb{Z}/5\mathbb{Z}$.

Let $\mathcal{O}_0 = \mathbb{Z}[\sqrt{-47}]$ be the order of conductor 2, $\text{Cl}(\mathcal{O}_0) \cong \mathbb{Z}/5\mathbb{Z}$.

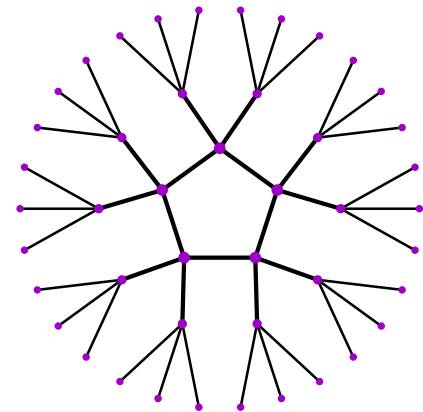
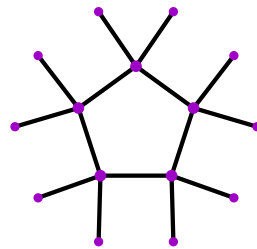
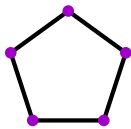
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The prime $\ell = 3$ splits in K : $(-47/3) = 1$. How does it factor in $\mathcal{O}_0$?

$$3\mathcal{O}_0 = \underbrace{(3, \sqrt{-47} - 1)}_{=\mathcal{L}} \underbrace{(3, \sqrt{-47} + 1)}_{=\overline{\mathcal{L}}}, \quad \text{ord}([\mathcal{L}]) = 5 \text{ in } \text{Cl}(\mathcal{O}_0)$$



$$V_0 = S_5$$

Strategy I

What do we *need* for (V_0, ℓ, d) to appear in $\mathcal{G}_\ell(\mathbb{F}_{p^k})$?

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- ▶ Crater (V_0) : need an order $\mathcal{O}_0$ such that a prime $\mathcal{O}_0$ -ideal lying above ℓ corresponds to V_0 .

Strategy I

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Lemma. If $p = \mathfrak{p}\bar{\mathfrak{p}}$ splits in K , $\mathcal{O}$ an order with conductor coprime to p , then there exists $E/\bar{\mathbb{F}}_p$ with $\text{End}(E) = \mathcal{O}$ and

$$j(E) \in \mathbb{F}_{p^k} \iff \text{ord}(\underbrace{[\mathfrak{p} \cap \mathcal{O}]}_{\in \text{Cl}(\mathcal{O})}) \mid k.$$

Strategy II

Fix K with discriminant < -4 , $\mathcal{O}_0$ an order of conductor c coprime to ℓ . Let V_0 be the crater type for the behaviour of ℓ in $\mathcal{O}_0$.

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Theorem (GGK). Consider the volcano $V = (V_0, \ell, d)$. Fix $k \geq 1$. Define the sets

$$X = \{\sigma \in \text{Cl}(\mathcal{O}_{d+1}) : \text{ord}(\sigma) \nmid k \text{ and } \text{ord}(\pi_d(\sigma)) \mid k\},$$

$$\mathcal{P} = \{p \text{ prime} : V \text{ appears in } \mathcal{G}_\ell(\mathbb{F}_{p^k}) \text{ with crater order } \mathcal{O}_0\}.$$

Then

- ▶ $X = \emptyset \implies \mathcal{P} = \emptyset$.
- ▶ $X \neq \emptyset \implies \mathcal{P}$ has positive density in the primes.

Strategy III

Proof Idea. There exists a finite abelian extension H_d of K such that $\text{Gal}(H_d/K) \cong \text{Cl}(\mathcal{O}_d)$ (this is the **ring class field** of $\mathcal{O}_d$). As X is closed under conjugation, we can use the Chebotarev density theorem.

Strategy III

Proof Idea. There exists a finite abelian extension H_d of K such that $\text{Gal}(H_d/K) \cong \text{Cl}(\mathcal{O}_d)$ (this is the **ring class field** of $\mathcal{O}_d$). As X is closed under conjugation, we can use the Chebotarev density theorem.

Corollary. Fix a volcano $V = (V_0, \ell, d)$ and $k \geq 1$.

- (a) If there exists *an* $\mathcal{O}_0$ compatible with (V_0, ℓ) such that $X \neq \emptyset$, then V has a positive-density set of k -explosive primes.
- (b) If $X = \emptyset$ for *all* $\mathcal{O}_0$ compatible with (V_0, ℓ) , then V has **no** k -explosive primes.

Sample case: crater is easy

Theorem (GGK). Let $V_0 = I_1, S_n$, $\ell \geq 3$, $d \geq 1$, and $n \geq 1$. If $v_\ell(k) < d$, then (V_0, ℓ, d) has (infinitely many) k -explosive primes.

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Theorem (GGK). Let $V_0 = I_1, S_n$, $\ell \geq 3$, $d \geq 1$, and $n \geq 1$. If $v_\ell(k) < d$, then (V_0, ℓ, d) has (infinitely many) k -explosive primes.

Proof.

- ▶ Crater (V_0): $V_0 = I_1$ easy. $V_0 = S_n$ less easy, but BCP already did it. Any order $\mathcal{O}_0$ will do!
- ▶ Depth (d): ...

Sample case: depth is more involved

$$X = \left\{ \sigma \in \text{Cl}(\mathcal{O}_{d+1}) \quad : \begin{array}{l} \text{ord}(\sigma) \nmid k, \\ \text{ord}(\pi_d(\sigma)) \mid k \end{array} \right\}$$

$$\begin{array}{c} \text{Cl}(\mathcal{O}_{d+1}) \\ \downarrow \pi_d \\ \text{Cl}(\mathcal{O}_d) \end{array}$$

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$$\begin{array}{ccccccc}
 & & 1 & & & & \\
 & & \downarrow & & & & \\
 & & \ker \pi_d & & & & \\
 & & \downarrow & & & & \\
 1 & \longrightarrow & K_{d+1} & \longrightarrow & \mathrm{Cl}(\mathcal{O}_{d+1})[\ell^\infty] & \longrightarrow & \mathrm{Cl}(\mathcal{O}_0)[\ell^\infty] \longrightarrow 1 \\
 & & & & \downarrow \pi_d & & \parallel \\
 1 & \longrightarrow & K_d & \longrightarrow & \mathrm{Cl}(\mathcal{O}_d)[\ell^\infty] & \longrightarrow & \mathrm{Cl}(\mathcal{O}_0)[\ell^\infty] \longrightarrow 1 \\
 & & & & \downarrow & & \\
 & & & & 1 & &
 \end{array}$$

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$$\begin{array}{ccc} 1 & & 1 \\ \downarrow & & \downarrow \\ \ker \pi'_d & \stackrel{=}{=} & \ker \pi_d \end{array}$$

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$$\begin{array}{ccccccc} 1 & \longrightarrow & \kappa_{d+1} & \longrightarrow & \text{Cl}(\mathcal{O}_{d+1})[\ell^\infty] & \longrightarrow & \text{Cl}(\mathcal{O}_0)[\ell^\infty] \longrightarrow 1 \\ & & \downarrow \pi'_d & & \downarrow \pi_d & & \parallel \\ 1 & \longrightarrow & \kappa_d & \longrightarrow & \text{Cl}(\mathcal{O}_d)[\ell^\infty] & \longrightarrow & \text{Cl}(\mathcal{O}_0)[\ell^\infty] \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \\ & & 1 & & 1 & & \end{array}$$

$$\ker \pi'_d \cap \ell^{\nu_\ell(k)} \kappa_{d+1} \neq \{0\} \implies X' \neq \emptyset \iff X \neq \emptyset \implies \star$$

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$$\begin{array}{ccc} 1 & & 1 \\ \downarrow & & \downarrow \\ \langle \ell^{d-1} \rangle & \xlongequal{\quad} & \ker \pi_d \end{array}$$

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$$v_\ell(k) < d \implies \langle \ell^{d-1} \rangle \cap \ell^{v_\ell(k)}(\mathbb{Z}/\ell^d\mathbb{Z}) \neq \{0\} \implies X' \neq \emptyset \iff X \neq \emptyset \implies \star$$

What next?

- ▶ Can we apply these ideas to higher dimensional abelian varieties?
- ▶ Can we find similarly nice structure in the isogeny graphs of higher dimensional abelian varieties?

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Thank you!